Locally Robust Bayesian Inference for State Space Models

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Summary

- Bayesian inference on a finite-dimensional parameter proceeds by updating the prior on the basis of information in the likelihood of the data

- Inference results are normally insensitive to the choice of prior (BvM theorem) but sensitive to even slight misspecification in the likelihood

- We want to:
  1) Robustify Bayesian inference against local misspecification in the likelihood (U. Müller and Norets (2024))

  2) Through a tractable manner that involves minimal alterations to the underlying model (this paper)

- It turns out that, for a wide variety of state space models, both goals can be achieved

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Motivation

- Consider estimating a simple AR(1) model

$$y_t = \phi y_{t-1} + \epsilon_t, \quad \mathbb{E}[\epsilon_t \mid \mathcal{F}_{t-1}] = 0$$

- The AR parameter ϕ is identified through the conditional moment equality

$$\mathbb{E}_{t-1}[y_t - \phi y_{t-1}] = 0,$$

  and frequentist inference on ϕ can proceed by making use of this condition (e.g. GMM)

- To conduct Bayesian inference on ϕ, what we normally do is impose a functional form on the distribution of ϵ_t and then use Bayes rule to obtain the posterior distribution of ϕ

- When are posterior results robust to likelihood misspecification?

Case 1: Normal Likelihood

- Suppose we use the Gaussian likelihood

$$y_t \mid \mathcal{Y}_{t-1} \sim \mathcal{N}\left[\phi y_{t-1},\ \sigma^2\right].$$

- Under the BvM theorem, under any well-behaved prior the posterior distribution of ϕ converges in total variation distance to

$$\mathcal{N}\left[\hat{\phi}_T^{MLE},\ I_0^{-1}\right],$$

  where $\hat{\phi}_T^{MLE}$ is the normal MLE of ϕ and I_0 the associated Fisher information.

- We know that $\hat{\phi}_T^{MLE}$ is the OLS estimator of ϕ and that it is asymptotically normal under the MDS assumption regardless of the DGP of ϵ_t—So posterior inference is robust to likelihood misspecification!

Case 2: A Model with Stochastic Volatility

- Now we want to model time-varying volatility: in the usual Bayesian approach to stochastic volatility, we estimate the model

$$y_t = \phi y_{t-1} + e^{\frac{1}{2}\omega_t}\epsilon_t, \quad \omega_{t+1} = \beta\omega_t + \sigma_\omega v_{t+1},$$
$$(\epsilon_t, v_{t+1})' \overset{iid}{\sim} \mathcal{N}\left[O_{2\times 1},\ I_2\right].$$

- Suppose the true model contains a leverage effect:

$$y_t = \phi y_{t-1} + e^{\frac{1}{2}\omega_t}\epsilon_t, \quad \omega_{t+1} = \beta\omega_t + \sigma_\omega v_{t+1},$$
$$(\epsilon_t, v_{t+1})' \overset{iid}{\sim} \mathcal{N}\left[O_{2\times 1},\ \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right].$$

- As long as $\rho \neq 0$, the MLE ϕ for $\hat{\phi}_T^{MLE}$ is asymptotically biased for ϕ

Case 2: A Model with Stochastic Volatility

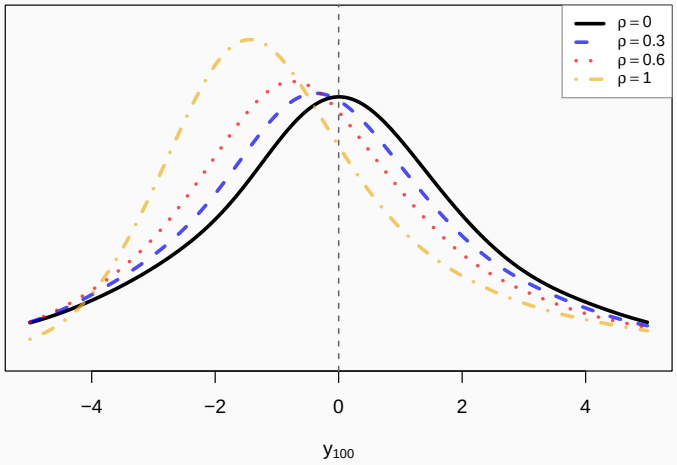


Figure: Distribution of y_{100}

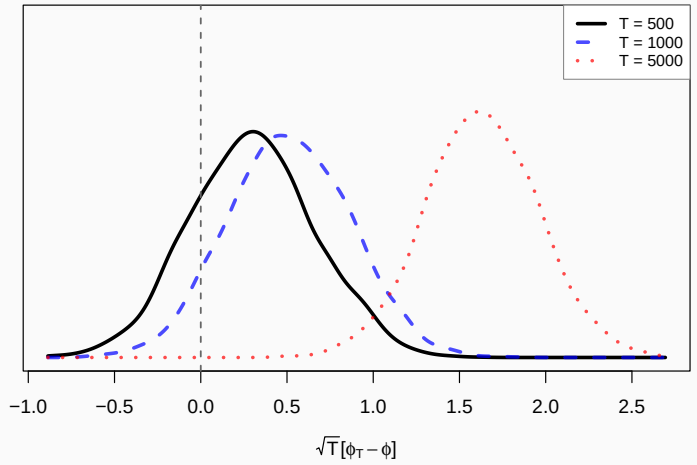


Figure: Posterior of ϕ

The data is generated using parameter values: $\phi = 0.9$, $\beta = 0.95$, and $\sigma_\omega^2 = 0.2$, $\rho = 0.9$

- The posterior draws of $\sqrt{T}(\phi - 0.9)$ are obtained from the baseline model with no leverage
- The posterior is not $\sqrt{T}$-stable; in fact, it is inconsistent

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Traditional Bayesian Approaches to Likelihood Misspecification

- The classical Bayesian solution to robustify inference against likelihood misspecification to treat the unknown likelihood as an infinite-dimensional parameter $\eta \in H$ and impose a prior on (θ, η)

- (Bickel and Kleijn (2012), The Semiparametric BvM Theorem) If:
  i) If the prior over H contracts toward the truth at a suitable rate that is slower than $\sqrt{T}$, and
  ii) The posterior of (θ, η) is consistent,

  then the marginal posterior of $\sqrt{T}[\theta - \theta_0]$ converges to the normal distribution centered at the semiparametrically efficient estimator $\hat{\theta}_T$

- The assumptions of the theorem are tricky and have to be verified on a model-by-model basis

An Alternative Approach to Likelihood Misspecification

- The semiparametric BvM imposes conditions on the prior that pushes the posterior of θ to its target; another approach is to focus on adjusting the likelihood

- Two main approaches to likelihood adjustment in the statistical literature:
  1) (Andrade and O'Hagan (2006); Lopes and Tobias (2011)) Embed the baseline likelihood in a more general class of models, usually a class of mixture distributions with heavy tails

  2) (Greco, Racugno, and Ventura (2008); Ruli, Sartori, and Ventura (2020)) Conduct inference using robust pseudo-likelihood whose MLE is robust to misspecification in the frequentist sense

- U. Müller and Norets (2024) refines previous approaches by combining with theory on semiparametric efficiency

Forming the Augmented Model

- Let θ be the parameter of interest and η a possibly infinite dimensional parameter
- (Van Der Vaart and Wellner (2021)) We should be able to introduce additional parameters δ that point in the direction of the projection of the baseline score onto η to "soak up" the first order effects η has on estimating θ
- Consider an augmented likelihood of the form

$$q(y \mid \theta, \delta) = c(\theta, \delta) k(y \mid \theta, \delta) p(y \mid \theta)$$

  where $p(y \mid \theta)$ is the likelihood of the original model and

$$\frac{\partial \log q(y \mid \theta, \delta)}{\partial \theta} = s(y \mid \theta), \quad \frac{\partial \log q(y \mid \theta, \delta)}{\partial \delta} = \Pi_\eta s(y \mid \theta)$$

- The score with respect to δ is precisely the direction of the least favorable submodel
- Shown that posterior of θ derived from augmented likelihood is equivalent in the limit to the distribution of a frequentist estimator that is robust to local misspecification

Limitations with Existing Approaches

- Difficult to construct pseudo/augmented likelihood for posterior sampling; in particular, unknown normalization constant $c(\theta, \delta)$ depends on parameters we want to sample

- Another complication: U. Müller and Norets (2024) only deals with iid data, but extending semiparametric theory to non-iid data is complicated due to loss of inner product space structure on tangent space

- This paper:
  1) Proposes minimal working assumptions under which an efficient locally robust estimator exists in a time series setting given conditional (or unconditional) moment equations

  2) Derives easy-to-sample forms of the augmented likelihood for a wide class of state space models

Summary of Related Literature

Bayesian/Frequentist Asymptotic Equivalence Kleijn and van der Vaart (2012) (Parametric BvM); Bickel and Kleijn (2012) (Semiparametric BvM)

Likelihood Robustness Greco, Racugno, and Ventura (2008) (Pseudo/Augmented likelihood); Andrade and O'Hagan (2006) (Embedding likelihood)

Semiparametric Estimation Bickel, Klaassen, et al. (1993) (General semiparametric theory) ; Newey (1990) (Semiparameric efficiency bounds) ; Chernozhukov et al. (2018) (Neyman orthogonalization)

Semiparametrics in Time Series Chamberlain (1987) (Efficiency bounds for time series) ; Drost, Klaassen, and Werker (1997) (Adaptive Estimation); Greenwood, U. U. Müller, and Wefelmeyer (2004) (Markov processes) ; Fiorentini and Sentana (2020) (Embedding within normal mixtures)

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Semiparametric Estimation Theory

- Our setup:
  1) **Parameters**
     A finite-dimensional paramter $\theta \in \Theta$ of interest and a possibly infinite dimensional paramter $\eta \in H$

  2) **DGP**
     A stochastic process $\{y_t\}_{t\in\mathbb{Z}}$ specified by the family of conditional one-step forward densities $\{f_t(\cdot \mid \mathcal{Y}_{t-1}; \theta, \eta) \mid \theta \in \Theta, \eta \in H\}$

  3) **The Objective**
     Conduct inference on θ under the baseline (not necessarily the truth!) model $\eta_0 \in H$, but also want inference to be robust to small misspecifications in the nuisance parameter

- Let $\dot{l}_t(y_t)$ be the baseline time t score increment of the model evaluated at $\theta = \theta_0$ (true parameter values) and $\eta = \eta_0$ (the baseline model)

- Time series case is straightforward generalization of iid case under conditional/unconditional moment conditions

Semiparametric Estimation Theory: Table of Differences

	IID	Non-IID
Local Deviations	Paths $r \mapsto \eta_r$ satisfying DQM condition with score function $g(\cdot)$	Paths $r \mapsto \eta_r$ satisfying ▸ uniform DQM condition with score function sequence $\{g_t(y^t)\}_{t\in\mathbb{N}}$
Identification Condition	Moment condition $\mathbb{E}^{\eta_r}[\psi(y_i;\theta)] = 0$ for all $r \geq 0$	Conditional moment condition $\mathbb{E}^{\eta_r}[\psi_t(y^t;\theta) \mid \mathcal{Y}_{t-1}] = 0$ for all $r \geq 0, t \geq 0$
Tangent Space $\eta\dot{\mathcal{P}}_P$	Score functions $g(\cdot)$ satisfying DQM and orthogonality condition $\mathbb{E}^{\eta_0}[\psi(y_i;\theta_0)g(y_i)] = 0$	Score sequences $\{g_t(y^t)\}$ satisfying uniform DQM and ▸ orthogonality condition $\mathbb{E}^{\eta_0}[\psi_t(y^t;\theta_0)g_t(y^t) \mid \mathcal{Y}_{t-1}] = 0$
Orthogonalization	Projection of baseline score $\dot{l}(\cdot)$ onto orthogonal complement of $\eta\dot{\mathcal{P}}_P$ $\Rightarrow$ Projection of $\dot{l}(\cdot)$ onto span of $\{\psi(\cdot;\theta_0)\}$	Projection of each $\dot{l}_t(\cdot)$ onto span of $\{1, \psi_t(y^t;\theta_0)\}$ $\Rightarrow$ Projection onto span of $\{\psi_t(y^t;\theta_0)\}$ by MDS property
Efficient Score	$\tilde{l}(y_i) = \left(\mathbb{E}^{\eta_0}\left[\frac{\partial\psi(y_i;\theta_0)}{\partial\theta}\right]\right)' \times \left(\mathbb{E}^{\eta_0}\left[\psi(y_i;\theta_0)\psi(y_i;\theta_0)'\right]\right)^{-1} \times \psi(y_i;\theta_0)$	$\tilde{l}_t(y_t) = \left(\mathbb{E}^{\eta_0}\left[\frac{\partial\psi_t(y^t;\theta_0)}{\partial\theta} \mid \mathcal{Y}_{t-1}\right]\right)' \times \left(\mathbb{E}^{\eta_0}\left[\psi_t(y^t;\theta_0)\psi_t(y^t;\theta_0)' \mid \mathcal{Y}_{t-1}\right]\right)^{-1} \times \psi_t(y^t;\theta_0)$

Semiparametric Estimation Theory: Locally Robust Estimator

- Locally robust estimator will be

$$\sqrt{T}\left[\hat{\theta}_T^{LR} - \theta_0\right] = \tilde{I}_0^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^{T} \tilde{l}_t(y_t) + o_p(1)$$

  where

$$\tilde{I}_0 = \operatorname*{plim}_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \Lambda_t(\theta_0)' V_t(\theta_0)^{-1} \Lambda_t(\theta_0)$$

- We can show $\hat{\theta}_T^{LR}$ converges to $\mathcal{N}\left[O_{k\times 1},\ \tilde{I}_0^{-1}\right]$ under identified deviations

- Can also show that the asymptotic distribution of any locally robust estimator must be a convolution involving $\mathcal{N}\left[O_{k\times 1},\ \tilde{I}_0^{-1}\right]$

  $\Rightarrow \hat{\theta}_T^{LR}$ is the locally robust estimator of θ that minimizes the efficiency loss from relying on the identification condition as our only source of information

Semiparametric Estimation Theory: Locally Robust Estimator

Theorem 1 (Local Robustness and Efficiency)

Let $r \mapsto \eta_r$ be an identified deviation that satisfies the uniform DQM with score function sequence $\{g_t(y^t)\} \in \eta\dot{\mathcal{P}}_P$ admitting positive asymptotic variance, and $h \in \mathbb{R}^k$. Then,

$$\sqrt{T}\left[\hat{\theta}_T^{LR} - \left(\theta_0 + \frac{h}{\sqrt{T}}\right)\right] \overset{\theta_0+h/\sqrt{T},\eta_{1/\sqrt{T}}}{\Rightarrow} \mathcal{N}\left[O_{k\times 1},\ \tilde{I}_0^{-1}\right].$$

Let $\mathcal{T}_T$ be a regular estimator of θ:

$$\sqrt{T}\left[\mathcal{T}_T - \left(\theta_0 + \frac{h}{\sqrt{T}}\right)\right] \overset{\theta_0+h/\sqrt{T},\eta_{1/\sqrt{T}}}{\Rightarrow} \mathcal{T}^*$$

for any $h \in \mathbb{R}^k$ and identified deviation that satisfies the uniform DQM. Then, there exists a probability distribution M such that

$$\mathcal{T}^* = \mathcal{N}\left[O_{k\times 1},\ \tilde{I}_0^{-1}\right] * M.$$

MLE in the AR-SV Model

Model : $$y_t = \phi y_{t-1} + e^{\frac{1}{2}\omega_t}\epsilon_t, \quad \omega_t = \beta\omega_{t-1} + \sigma_\omega v_t, \quad (\epsilon_t, v_t)' \overset{iid}{\sim} \mathcal{N}\left[O_{2\times 1},\ I_2\right]$$

- The baseline score for ϕ is

$$\dot{l}_t(y_t) = \mathbb{E}_t\left[e^{-\omega_t}\right] y_{t-1}\left(y_t - \phi_0 y_{t-1}\right),$$

  where $\mathbb{E}_t\left[e^{-\omega_t}\right]$ is a highly non-linear function of y_t and past values of y

- In order for the MLE to be asymptotically unbiased under a local deviation $r \mapsto \eta_r$ with score sequence $\{g_t(y^t)\}_{t\in\mathbb{N}}$, we require

$$\mathbb{E}\left[\dot{l}_t(y_t) g_t(y^t)\right] = O_{k\times 1},$$

  or for $g_t(y^t)$ to be orthogonal to a highly non-linear function of the data up to time t
  $\Rightarrow$ Does not hold in general

Locally Robust Estimation in the AR-SV Model

$$\text{Model}: \quad y_t = \phi y_{t-1} + e^{\frac{1}{2}\omega_t}\epsilon_t, \quad \omega_t = \beta\omega_{t-1} + \sigma_\omega v_t, \quad (\epsilon_t, v_t)' \overset{iid}{\sim} \mathcal{N}\left[O_{2\times 1},\ I_2\right]$$

$$\text{Baseline Score}: \quad \dot{l}_t(y_t) = \mathbb{E}_t\left[e^{-\omega_t}\right] y_{t-1}\left(y_t - \phi_0 y_{t-1}\right)$$

- In this model, the conditional mean equation and efficient score are

$$\mathbb{E}_{t-1}\left[y_t - \phi y_{t-1}\right] = 0, \quad \tilde{l}_t(y_t) = \frac{1}{\mathbb{E}_{t-1}\left[e^{\omega_t}\right]} y_{t-1}\left(y_t - \phi_0 y_{t-1}\right)$$

- By Jensen's inequality,

$$\mathbb{E}_{t-1}\left[\dot{l}_t(y_t)^2\right] = \mathbb{E}_{t-1}\left[e^{-\omega_t}\right] y_{t-1}^2 \geq \frac{y_{t-1}^2}{\mathbb{E}_{t-1}\left[e^{\omega_t}\right]} = \mathbb{E}_{t-1}\left[\tilde{l}_t(y_t)^2\right]$$

- Higher conditional variance in ω_t
  $\Rightarrow$ Greater difference due to Jensen's inequality
  $\Rightarrow$ More information gain from working with parametric model

Locally Robust Estimation using Unconditional Moment Conditions

- Relying only on conditional mean conditions preclude estimation of many important models: consider the MA(1) model

$$y_t = \epsilon_t + \alpha\epsilon_{t-1}, \quad \epsilon_t \overset{WN}{\sim} (0, \sigma^2)$$

- The MA parameter α is identified via the unconditional moment equation

$$\mathbb{E}\left[y_t y_{t-1} - \frac{\alpha}{1+\alpha^2} y_{t-1}^2\right] = 0$$

  $+$ invertibility (or non-invertibility) assumptions on α

- There is **no** conditional moment equation that allows us to identify α when we do not have recourse to data in the infinite past

- Can develop semiparametric estimation theory under unconditional moment conditions under additional assumptions on weak dependence

Locally Robust Estimation using Unconditional Moment Conditions

- Interested in identified deviations $r \mapsto \eta_r$ that satisfy

$$\mathbb{E}^{\eta_r}\left[m(y_t, \cdots, y_{t-\tau}; \theta)\right] = O_{q\times 1} \quad \forall r \geq 0$$

for some fixed $\tau > 0$. Define $z_t = (y_t, \cdots, y_{t-\tau})$.

- ▸ Orthogonality condition for score sequences $\{g_t(y^t)\}_{t\in\mathbb{N}}$ is

$$\mathbb{E}^{\eta_0}\left[m(z_t; \theta_0) g_t(y^t)\right] = O_{q\times 1}.$$

Efficient score $\tilde{l}_t(\cdot)$ constructed by "projection" onto $\{1, m(z_t; \theta_0)\}$:

$$\begin{aligned}\tilde{l}_t(y_t) = &\operatorname{Cov}^{\eta_0}\left[\dot{l}_t(y_t), m(z_t; \theta_0) \mid \mathcal{Y}_{t-1}\right] \\ &\times \left(\operatorname{Var}^{\eta_0}\left[m(z_t; \theta_0) \mid \mathcal{Y}_{t-1}\right]\right)^{-1} \left(m(z_t; \theta_0) - \mathbb{E}^{\eta_0}\left[m(z_t; \theta_0) \mid \mathcal{Y}_{t-1}\right]\right)\end{aligned}$$

- Same ▸ local robustness and efficiency result holds if we restrict attention to score sequences $\{g_t(y^t)\}_{t\in\mathbb{N}}$ such that 1. orthogonality condition is satisfied and 2. $\{m(z_t; \theta_0) g_t(y^t)\}_{t\in\mathbb{N}}$ is an L^1-mixingale

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Robustifying Bayesian Inference

- Now augment baseline model with additional parameter δ: denote the augmented conditional density $\mathcal{A}_t(y_t \mid \mathcal{Y}_{t-1}; \theta, \delta)$

- We generalize the analogous result in U. Müller and Norets (2024):

Theorem 2 (Local Robustenss for the Augmented Model)

Suppose the augmented conditional density $\mathcal{A}_t(y_t \mid \mathcal{Y}_{t-1}; \theta, \delta)$ satisfies the derivative conditions

$$\frac{\partial \log \mathcal{A}_t(y_t \mid \mathcal{Y}_{t-1}; \theta, \delta)}{\partial \theta} |_{\theta=\theta_0, \delta=0} = \dot{l}_t(y_t)$$
$$\frac{\partial \log \mathcal{A}_t(y_t \mid \mathcal{Y}_{t-1}; \theta, \delta)}{\partial \delta} |_{\theta=\theta_0, \delta=0} = \dot{l}_t(y_t) - \tilde{l}_t(y_t).$$

Under a prior π on $\Theta \times \mathbb{R}^k$ that has a continuous and positive Lebesgue density at $(\theta_0, O_{k\times 1})$, if

$$\mathbb{P}\left(\left| \sqrt{T}\left[\theta - \theta_0\right] \right| \vee \left| \sqrt{T}\delta \right| > M_T \mid X^T \right) = o_p(1)$$

for any $M_T \nearrow +\infty$,

$$\left\| \Pi^{Aug}_{\sqrt{T}[\theta-\theta_0]} \left[\cdot \mid Y^T \right] - \mathcal{N}\left[\cdot \mid \sqrt{T}\left[\hat{\theta}^{LR}_T - \theta_0 \right],\ \tilde{I}_0^{-1} \right] \right\|_{TV} = o_{p_{\eta_{1/\sqrt{T}}}}(1)$$

for any local deviation $r \mapsto \eta_r$ that is continguous with respect to the baseline model.

Robustifying Bayesian Inference

- The general approach to constructing the augmented conditional density: start by constructing the intermediate density $\mathcal{A}_t^{int}(y_t \mid \mathcal{Y}_{t-1}; \theta, \delta)$ satisfying

$$\frac{\partial \log \mathcal{A}_t^{int}(y_t \mid \mathcal{Y}_{t-1}; \theta, \delta)}{\partial \theta}|_{\theta=\theta_0, \delta=0} = \dot{l}_t(y_t)$$
$$\frac{\partial \log \mathcal{A}_t^{int}(y_t \mid \mathcal{Y}_{t-1}; \theta, \delta)}{\partial \delta}|_{\theta=\theta_0, \delta=0} = \tilde{l}_t(y_t),$$

  and then define

$$\mathcal{A}_t(y_t \mid \mathcal{Y}_{t-1}; \theta, \delta) = \mathcal{A}_t^{int}(y_t \mid \mathcal{Y}_{t-1}; \theta + \delta, -\delta)$$

- The intermediate density is an exponential tilt of the original density:

$$\mathcal{A}_t^{int}(y_t \mid \mathcal{Y}_{t-1}; \theta, \delta) = f_t(y_t \mid \mathcal{Y}_{t-1}; \theta) e^{\delta \tilde{l}_t(y_t)}.$$

- In general, the normalizing constant $\mathbb{E}\left[e^{\delta \tilde{l}_t(y_t)} \mid \mathcal{Y}_{t-1}\right]$ for the exponential tilt is unknown except for a few cases (e.g. a few distributions in the exponential family)

Robustifying Bayesian Inference

- We find that, if:
  1) We are interested in inference about a parameter that is identified via first or second moments, and
  2) The baseline model is a mixture normal distribution,

  the normalizing constant is known

- In a sense, we combine the two previous appraoches to likelihood robustification: embedding the baseline model in a flexible normal mixture setting to construct an augmented (or pseudo) likelihood

- So what's so special about mixture normals?

What's So Special about Mixture Normals?

- For simplicity, consider inference on the mean and variance of an iid process $\{y_t\}_{t \in N_+}$ whose distribution is an unrestricted finite normal mixture with known variances:

$$f(y_t) = \sum_{j=1}^{k} p_j \mathcal{N}\left[y_t \mid \mu_j,\ \sigma_j^2\right]$$

- The parameters of interest μ is defined as $\mu = \sum_{j=1}^{k} p_j \mu_j$ and identified via the first moment condition $\mathbb{E}[y_t - \mu] = 0$

- Fiorentini and Sentana (2020) show that the MLE of μ is equal to the first sample moment:

$$\hat{\mu}_T := \sum_{j=1}^{k} \hat{p}_{jT} \hat{\mu}_{jT} = \frac{1}{T} \sum_{t=1}^{T} y_t$$

$\Rightarrow$ The score of μ in the baseline model must be exactly the efficient score $\tilde{l}_t(y_t) = \frac{y_t - \mu}{\sigma^2}$!

What's So Special about Mixture Normals?

- Since μ is a smooth function of the primitive parameters p_j, μ_j, σ_j^2, the score of μ must be a linear combinations of the scores of p_j, μ_j, σ_j^2
  $\Rightarrow$ The efficient score is a linear combination of the components of the baseline score

- Letting this linear combination be

$$\frac{y_t - \mu}{\sigma^2} = \sum_{j=1}^{k} \left[a_j \frac{\partial \log f(y_t)}{\partial p_j} + b_j \frac{\partial \log f(y_t)}{\partial \mu_j} \right],$$

  the intermediate augmented model can be constructed by separately tilting the mixture weights and mixture means by a_j and b_j, respectively

- Computations reveal augmented density is

$$f^a(y_t; \delta) = \sum_{j=1}^{k} \frac{p_j e^{-\delta(\mu_j - \mu)/\sigma^2}}{\sum_{j'=1}^{k} p_{j'} e^{-\delta(\mu_{j'} - \mu)/\sigma^2}} \mathcal{N}\left[y_t \mid \mu_j + \delta\left(1 - \frac{\sigma_j^2}{\sigma^2}\right),\ \sigma_j^2 \right].$$

Extension Beyond Finite Mixtures

- Finite case easily generalized—to robustify inference for a mean parameter under any normal mixture model requires separate adjustment of weights and mixture component means

- Normalization constant $\sum_{j'=1}^{k} p_{j'} e^{-\delta(\mu_{j'}-\mu)/\sigma^2}$ cannot be easily computed when $k \to \infty$ and we are back to the problem of U. Müller and Norets (2024)

- Two straightforward cases (common in economic applications!) when computation is made possible:
  - When $\mu_j = \mu$ for all j, so that only adjustment of mixture component mean is required
    $\Rightarrow$ Scale mixture case, encompasses stochastic volatility models
  - When we know how to exponentially tilt weights $\{p_j\}_j$ (e.g. when they are normal)
    $\Rightarrow$ Linear location mixture case, encompasses the basic linear Gaussian state space models

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Example: Regression with t-Distributed Errors

- Consider the autoregessive regession model

$$y_t = c + \beta y_{t-1} + \epsilon_t, \quad \epsilon_t \overset{iid}{\sim} \mathbf{t}_v$$

- We can write the model as a scale mixture of normals:

$$f_t(y_t \mid \mathcal{Y}_{t-1}; c, \beta) = \int_0^\infty \mathcal{N}\left[y_t \mid c + \beta y_{t-1},\ S\right] p(S) dS, \quad p \sim \mathcal{IG}\left[\frac{v}{2}, \frac{v}{2}\right]$$

  where the conditional variance is $\sigma_t^2 = \mathbb{E}_{t-1}\left[(y_t - c - \beta y_{t-1})^2\right] = \frac{v}{v-2}$

- The augmented density is

$$\begin{aligned}
&\mathcal{A}_t(y_t \mid \mathcal{Y}_{t-1}; c, \beta, \delta_c, \delta_\beta) \\
&\quad = \int_0^\infty \mathcal{N}\left[y_t \mid c + \beta y_{t-1} + \left(1 - S\frac{v-2}{v}\right)\left(\delta_c + \delta_\beta y_{t-1}\right),\ S\right] p(S) dS
\end{aligned}$$

Example: Regression with t-Distributed Errors

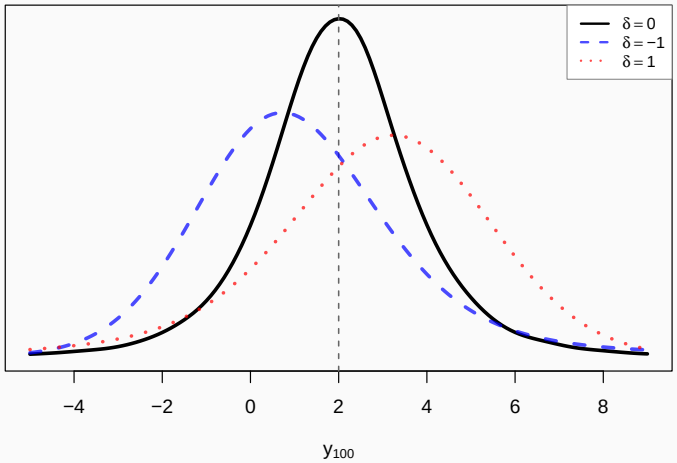


Figure: Least Favorable Submodel with $\delta_\beta = 0$

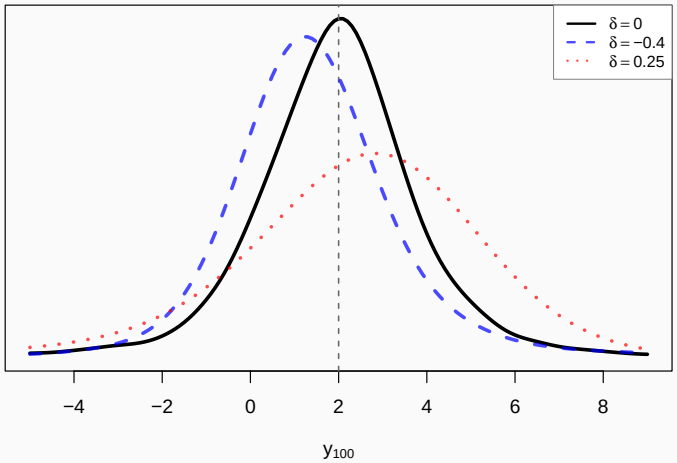


Figure: Least Favorable Submodel with $\delta_c = 0$

The data were generated under the parameter values $c = 1$, $\beta = 0.5$, $\nu = 2.5$

General Case: Stochastic Volatility Models

- State-space models of the form

$$y_t = g(\mathcal{Y}_{t-1};\theta) + \Sigma_t(\omega_t)^{\frac{1}{2}}\epsilon_t, \quad \omega_t \mid \omega_{t-1} \sim p(\omega_{t-1}), \quad \epsilon_t \overset{iid}{\sim} StdNormal$$

  are just scale normal mixtures with possibly continuous mixing variables

- Normal mixture representation is

$$f_t(y_t \mid \mathcal{Y}_{t-1};\theta) = \int_W \mathcal{N}\left[y_t \mid g(\mathcal{Y}_{t-1};\theta), \Sigma_t(\omega)\right] q_t(\omega;\theta) d\omega, \quad \omega_t \mid \mathcal{Y}_{t-1};\theta \sim q_t(\omega_t;\theta)$$

- If θ is identified via the conditional mean equation $\mathbb{E}_{t-1}\left[y_t - \mu_t(\theta)\right] = O_{m\times 1}$, formula for finite mixture case suggests augmented model satisfies

$$y_t \mid \omega_t;\theta,\delta \sim \mathcal{N}\left[g(\mathcal{Y}_{t-1};\theta+\delta) + \left(I_m - \Sigma_t(\omega_t)V_t(\theta_0)^{-1}\right)\frac{\partial g(\mathcal{Y}_{t-1};\theta)}{\partial\theta}\delta, \quad \Sigma_t(\omega_t)\right]$$

$$\omega_t \mid \mathcal{Y}_{t-1};\theta,\delta \sim q_t(\omega_t;\theta)$$

$$V_t(\theta) = \mathbb{E}_{t-1}\left[\Sigma_t(\omega_t);\theta\right]$$

General Case: Gaussian State Space Models

- For sampling purposes, the augmented model can be represented in terms of measurement + transition density instead of measurement + filtered density

Theorem 3 (Robust Inference for Scale Mixture of Normals)

Let the baseline model be the state-space model

$$y_t = g(\mathcal{Y}_{t-1};\theta) + \Sigma_t(\omega_t)^{\frac{1}{2}}\epsilon_t, \quad \omega_t \mid \omega_{t-1} \sim p(\omega_{t-1}), \quad \epsilon_t \overset{iid}{\sim} StdNormal,$$

and let θ be identified through the conditional mean equation

$$\mathbb{E}_{t-1}\left[y_t - g(\mathcal{Y}_{t-1};\theta)\right] = O_{m\times 1}.$$

Then, under the augmented model

$$y_t = g(\mathcal{Y}_{t-1};\theta+\delta) - \Sigma_t(\omega_t)\left(\mathbb{E}_{t-1}\left[\Sigma_t(\omega_t)\right]\right)^{-1}\frac{\partial g(\mathcal{Y}_{t-1};\theta)}{\partial\theta}\delta + \Sigma_t(\omega_t)^{\frac{1}{2}}\epsilon_t$$

$$\omega_t \mid \omega_{t-1} \sim p(\omega_{t-1})$$

$$\epsilon_t \overset{iid}{\sim} StdNormal,$$

posterior inference for θ is locally robust.

Example: Augmented AR-SV Model

$$\text{Model}: \quad y_t = \phi y_{t-1} + e^{\frac{1}{2}\omega_t}\epsilon_t, \quad \omega_t = \beta\omega_{t-1} + \sigma_\omega v_t, \quad (\epsilon_t, v_t)' \overset{iid}{\sim} \textit{StdNormal}$$

$$\text{Baseline Score}: \quad \dot{l}_t(y_t) = \mathbb{E}_t\left[e^{-\omega_t}\right] y_{t-1}\left(y_t - \phi_0 y_{t-1}\right)$$

$$\text{Efficient Score}: \quad \tilde{l}_t(y_t) = \frac{y_{t-1}}{\mathbb{E}_{t-1}\left[e^{\omega_t}\right]}\left(y_t - \phi_0 y_{t-1}\right)$$

- The augmented version of the AR-SV model is

$$y_t = \phi y_{t-1} + \left(1 - \frac{e^{\omega_t}}{\mathbb{E}_{t-1}\left[e^{\omega_t}\right]}\right)\delta y_{t-1} + e^{\frac{1}{2}\omega_t}\epsilon_t,$$

$$\omega_t = \beta\omega_{t-1} + \sigma_\omega v_t, \quad (\epsilon_t, v_t)' \overset{iid}{\sim} \textit{StdNormal}$$

- The augmented model is a version of the SV-in-mean model—if the truth is that volatility affects the mean equation, then the baseline model is erroneously precise

Example:Augmented AR-SV Model

- We evaluate the performance of the baseline AR-SV model with intercepts against its augmentation under the local alternative

$$y_t = \gamma + \phi y_{t-1} + \frac{c}{\sqrt{T}}\left(\omega_t - \frac{\mu}{1-\beta}\right) + e^{\frac{1}{2}\omega_t}\epsilon_t$$

$$\omega_t = \mu + \beta\omega_{t-1} + \sigma_\omega v_t$$

$$\begin{pmatrix}\epsilon_t \\ v_t\end{pmatrix} \overset{iid}{\sim} StdNormal.$$

- Generate 100 datasets for sample sizes $T = 250, 500$ under parameter values $\gamma = \mu = 0$, $\phi = 0.9$, $\beta = 0.9$, $\sigma_\omega = 0.5^2$ and $c = -1$

- Predictive quantities $\mathbb{E}_{t-1}\left[e^{\omega_t}\right]$ computed using the bootstrap particle filter

Example: Augmented AR-SV Model

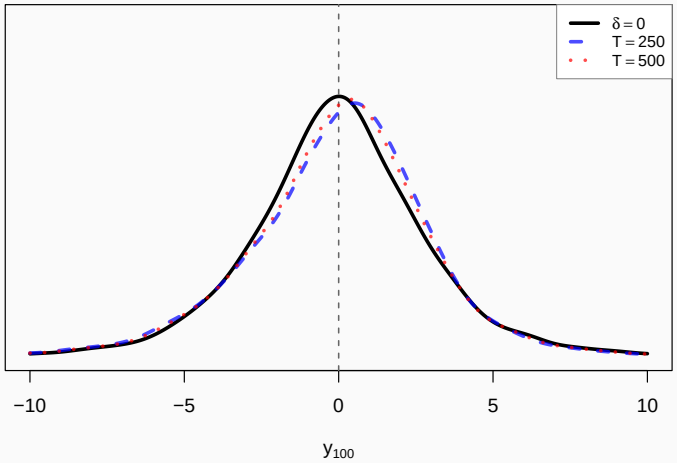


Figure: $c = -1$

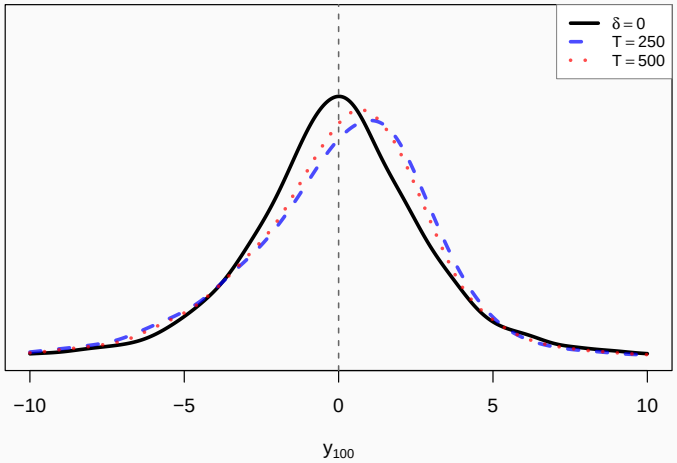


Figure: $c = -2$

- In both cases the local deviations beget only very small changes in the distribution
- The baseline model breaks down under either specification for c, while augmented model is robust

Example: Augmented AR-SV Model

	$\left\lvert\hat{\gamma}_T^{LR} - \mathbb{E}[\gamma \mid \mathcal{Y}_T]\right\rvert$		$\lvert\gamma - \mathbb{E}[\gamma \mid \mathcal{Y}_T]\rvert$		$\frac{sd(\gamma \mid \mathcal{Y}_T)}{sd(\hat{\gamma}_T^{LR})}$		Coverage Probability	
T	Baseline	Aug	Baseline	Aug	Baseline	Aug	Baseline	Aug
100	0.0627	0.0890	0.1234	0.1600	0.8965	1.6136	0.86	0.94
250	0.0347	0.0292	0.0681	0.0802	0.8513	1.2363	0.87	0.94
500	0.0263	0.0125	0.0519	0.0496	0.8386	1.1253	0.78	0.94

Table: Monte Carlo Averages for $c = -1$

	$\left\lvert\hat{\gamma}_T^{LR} - \mathbb{E}[\gamma \mid \mathcal{Y}_T]\right\rvert$		$\lvert\gamma - \mathbb{E}[\gamma \mid \mathcal{Y}_T]\rvert$		$\frac{sd(\gamma \mid \mathcal{Y}_T)}{sd(\hat{\gamma}_T^{LR})}$		Coverage Probability	
T	Baseline	Aug	Baseline	Aug	Baseline	Aug	Baseline	Aug
100	0.0417	0.2908	0.1016	0.2639	1.092	4.454	0.52	0.67
250	0.0313	0.0877	0.0679	0.0678	1.031	2.363	0.33	0.78
500	0.0268	0.0309	0.0634	0.0257	0.941	1.639	0.16	0.83

Table: Monte Carlo Averages for $c = -2$

Similar patterns for ▸ more general specifications

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When are Scale Mixtures Insufficient?

- Many models where latent factors ω_t appear in the mean equation directly (e.g. state space representation of ARMA, dynamic factor models)

- Main models of interest are linear Gaussian state space models

$$y_t = \Phi\omega_t + \Sigma^{\frac{1}{2}}\epsilon_t,$$
$$\omega_t = F\omega_{t-1} + \Omega^{\frac{1}{2}}v_t, \quad (\epsilon_t', v_t')' \overset{iid}{\sim} StdNormal,$$

  where parameter of interest Φ is identified via the unconditional moment equation

$$\mathbb{E}\left[\beta(z_{t-1};\Phi)\left(y_t - g(z_{t-1};\Phi)\right)\right] = O_{q\times 1}, \quad z_{t-1} = (y_{t-1}, \cdots, y_{t-\tau})$$

- From Kalman filter updating we know that $y_t \mid \omega_t$ and $\omega_t \mid \mathcal{Y}_{t-1}$ are both normal—we know how to adjust both mixture weights and mixture components

Robustification for Location Mixture of Normals

Theorem 4 (Robust Inference for Location Mixture of Normals)

Let the baseline model be the state-space model

$$y_t = \Phi\omega_t + \Sigma^{\frac{1}{2}}\epsilon_t,$$
$$\omega_t = F\omega_{t-1} + \Omega^{\frac{1}{2}}v_t, \quad (\epsilon_t', v_t')' \overset{iid}{\sim} StdNormal,$$

and let Φ be identified through the unconditional moment condition

$$\mathbb{E}[\beta(z_{t-1};\Phi)(y_t - g(z_{t-1};\Phi))] = O_{q\times 1}, \quad z_{t-1} = (y_{t-1}, \cdots, y_{t-\tau}).$$

Let $y_{t|t-1}(\theta)$ and $\Sigma_{t|t-1}(\theta)$ be the conditional expectation and variance of y_t given $\mathcal{Y}_{t-1}$ in the baseline model with parameters θ. Then, under the augmented model

$$y_t \mid \mathcal{Y}_{t-1};\theta,\delta \sim \mathcal{N}\left[y_{t|t-1}(\theta+\tilde{\delta}) - \Sigma_{t|t-1}(\theta)\Lambda_t(\theta)\delta,\ \Sigma_{t|t-1}(\theta+\tilde{\delta})\right],$$

where $\tilde{\delta}_i = 0$ if θ_i is not part of Φ and

$$\Lambda_t(\theta) = \beta(z_{t-1};\Phi)'\left[\beta(z_{t-1};\Phi)\Sigma_{t|t-1}(\theta)\beta(z_{t-1};\Phi)'\right]^{-1}\beta(z_{t-1};\Phi)\left(\frac{\partial g(z_{t-1};\Phi)}{\partial vec(\Phi)}\right),$$

posterior inference for Φ is locally robust.

Example: MA(1) Estimation

- Return to the MA(1) model

$$y_t = \epsilon_t + \alpha\epsilon_{t-1}, \quad \epsilon_t \overset{WN}{\sim} (0, \sigma^2),$$

where the mean parameter α is identified via the unconditional moment condition

$$\mathbb{E}[m(y_t, y_{t-1}; \alpha)] = 0, \quad m(y_t, y_{t-1}; \alpha) = y_t y_{t-1} - \frac{\alpha}{1+\alpha^2} y_{t-1}^2.$$

- The model can be cast in the following state-space form:

$$y_t = \underbrace{\begin{pmatrix} 1 & \alpha \end{pmatrix}}_{\Phi} w_t, \quad w_t = \underbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}_{F} w_{t-1} + \underbrace{\begin{pmatrix} \epsilon_t \\ 0 \end{pmatrix}}_{v_t}, \quad w_t = \begin{pmatrix} \epsilon_t \\ \epsilon_{t-1} \end{pmatrix}.$$

- Suppose the baseline model is the standard linear Gaussian state-space model where we assume $\epsilon_t \overset{iid}{\sim} \mathcal{N}\left[0, \sigma^2\right]$

Example: MA(1) Estimation

- Running the Kalman filter reveals that

$$y_1;\alpha \sim \mathcal{N}\left[0,\ \sigma^2(1+\alpha^2)\right], \quad y_t \mid \mathcal{Y}_{t-1};\alpha \sim \mathcal{N}\left[y_{t|t-1}(\alpha),\ \sigma^2_{t|t-1}(\alpha)\right]$$

- From the preceding theory the efficient score is

$$\begin{aligned}\tilde{l}_t(y_t) &= \Lambda_t(\alpha_0)\left[m(y_t, y_{t-1};\alpha_0) - \mathbb{E}_{t-1}\left[m(y_t, y_{t-1};\alpha_0)\right]\right] \\ &= \Lambda_t(\alpha_0) y_{t-1}\left[y_t - y_{t|t-1}(\alpha_0)\right] \\ &= \frac{\partial y_{t|t-1}(\alpha_0)}{\partial \alpha}\frac{1}{\sigma^2_{t|t-1}(\alpha_0)}\left[y_t - y_{t|t-1}(\alpha_0)\right]\end{aligned}$$

- The augmented model is given as

$$y_t \mid \mathcal{Y}_{t-1};\alpha,\delta \sim \mathcal{N}\left[y_{t|t-1}(\alpha+\delta) - \frac{\sigma^2_{t|t-1}(\alpha+\delta)}{\sigma^2_{t|t-1}(\alpha)}\frac{\partial y_{t|t-1}(\alpha)}{\partial \alpha}\delta,\ \sigma^2_{t|t-1}(\alpha+\delta)\right]$$

Example: MA(1) Estimation

- The baseline time t score increment is

$$\dot{l}_t(y_t) = \tilde{l}_t(y_t) + \frac{\partial \sigma^2_{t|t-1}(\alpha_0)}{\partial \alpha} \frac{1}{2\left(\sigma^2_{t|t-1}(\alpha_0)\right)^2} \left[\left(y_t - y_{t|t-1}(\alpha_0)\right)^2 - \sigma^2_{t|t-1}(\alpha_0)\right],$$

  and the properties of the normal distribution show us that

$$\mathbb{E}_{t-1}\left[\dot{l}_t(y_t)^2\right] = \mathbb{E}_{t-1}\left[\tilde{l}_t(y_t)^2\right] + \frac{1}{2\sigma^4_{t|t-1}(\alpha_0)} \left(\frac{\partial \sigma^2_{t|t-1}(\alpha_0)}{\partial \alpha}\right)^2$$

- Information gain from working with parametric Gaussian model is dependent on the shape of the mapping $\alpha \mapsto \sigma^2_{t|t-1}(\alpha)$

- Parametric Gaussian case is locally robust when we restrict our attention to the invertible case $|\alpha| < 1$
  $\Rightarrow$ But not the case when the model is non-invertible ($|\alpha| > 1$)

Example: Non-invertible MA(1) Estimation

- If $|\alpha| > 1$, then steady-state Kalman filtering yields, for large t,

$$\sigma^2_{t|t-1}(\alpha) = \sigma^2\alpha^2, \quad y_{t|t-1}(\alpha) = \frac{1}{\alpha}\left(y_{t-1} - y_{t-1|t-2}(\alpha)\right)$$

- Approximating with steady state values reveals

$$\frac{1}{2\sigma^4_{t|t-1}(\alpha_0)}\left(\frac{\partial\sigma^2_{t|t-1}(\alpha_0)}{\partial\alpha}\right)^2 \approx \frac{2}{\alpha^2},$$

  so information gain from assuming parametric Gaussian model is bounded below by a non-zero value

- Under the steady state approximation to the baseline and efficient scores we can also see that the Gaussian benchmark suffers from asymptotic bias under local alternatives

Example: Non-invertible MA(1) Estimation

- Consider the following local deviation from the baseline model:

$$y_t = \epsilon_t + \alpha\epsilon_{t-1}, \quad \epsilon_t \overset{iid}{\sim} \mathbf{t}_v, \quad v = \sqrt{T}/c$$

- Using the steady state approximation we can obtain the MLEs from the baseline and augmented models

- Compare distribution of $\sqrt{T}\left[\hat{\alpha}_T^{MLE} - \alpha\right]$ through simulations: use parameter values $\alpha = 1.1$ and $c = 2$

- Results show persisting bias in MLE under baseline model
  $\Rightarrow$ Augmentation should improve estimation

- Sidenote: similar bias appears in MLE of baseline Gaussian state space model for ARMA processes under invertibility assumptions

Example: Non-invertible MA(1) Estimation

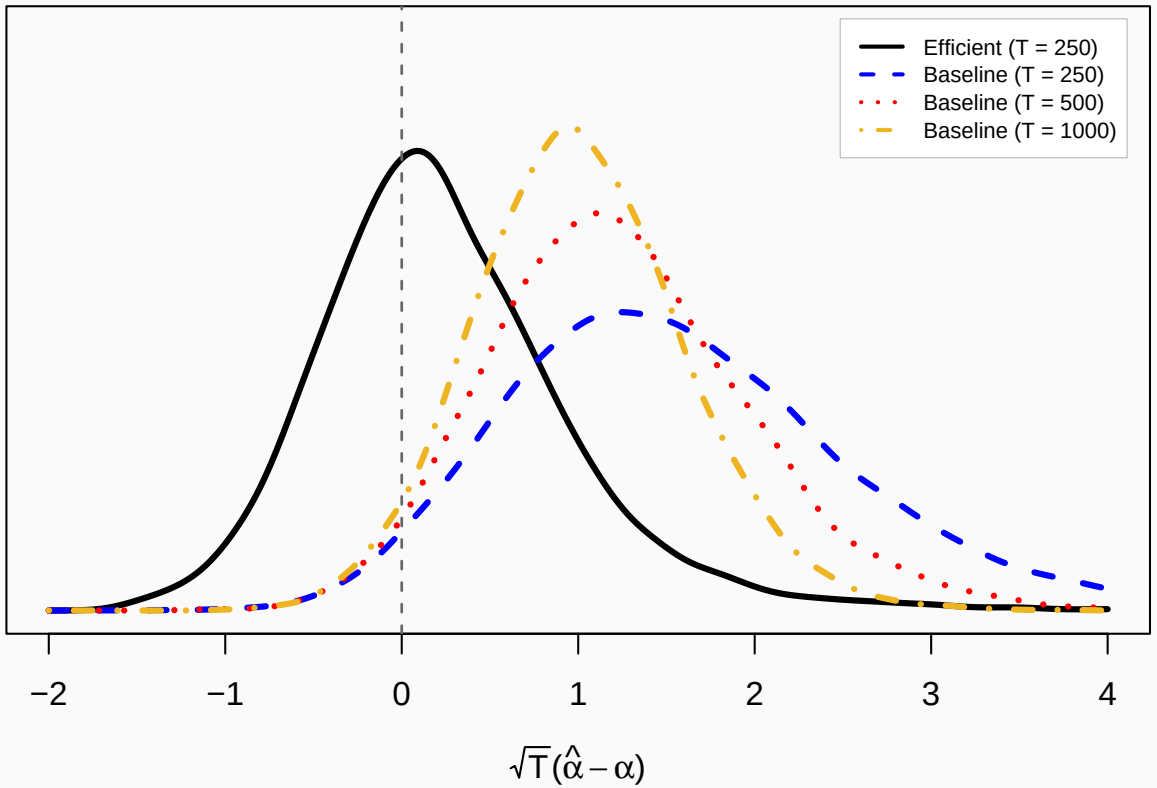


Figure: Asymptotic Distribution of MLEs for Baseline and Efficient Scores

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Conclusion

- We are interested in robustifying Bayesian inference to local misspecifications in the likelihood by augmenting the baseline likelihood
- When our parameter of interest is identified via first or second moments, and the baseline model is a finite normal mixture, we have a closed form for the augmented likelihood
- The finite mixture result can be generalized to scale and location mixtures of normals, allowing us to robustify inference on mean parameters in stochastic volatility models and linear Gaussian state space models
- Next steps: robustifying inference on mean parameters in state space models with latent factors in both mean and variance + robustifying inference on variance parameters

Semiparametric Estimation Theory: Local Deviations

Back

Local misspecification defined via one-dimensional paths $r \mapsto \eta_r$ in H that are locally asymptotically normal with respect to baseline

Definition 5 (Uniform DQM)

We say that a one-dimensional path $r \mapsto \eta_r$ satisfies the uniform DQM property with score sequence $\{g_t(y^t)\}_{t\in\mathbb{N}}$ if

$$\max_{1\le t\le T} \mathbb{E}\left|\sqrt{T}\left(\sqrt{\frac{f_t(y_t \mid \mathcal{Y}_{t-1};\theta_0,\eta_r)}{f_t(y_t \mid \mathcal{Y}_{t-1};\theta_0,\eta_0)}}-1\right)-\frac{1}{2}g_t(y^t)\right| = o(1) \quad \text{as } T\to\infty.$$

Theorem 6 (Uniform DQM implies LAN)

Let $r \mapsto \eta_r$ be a local deviation from η_0 that satisfies the uniform DQM property with score sequence $\{g_t(y^t)\}_{t\in N_+}$ such that $\frac{1}{T}\sum_{t=1}^{T} g_t(y^t)$ has a positive probability limit $\mathcal{I}(g)$. Then, $\{g_t(y^t)\}_{t\in\mathbb{N}}$ is an MDS and we have the LAN expansion

$$\log\prod_{t=1}^{T}\frac{f_t(Y_t \mid \mathcal{Y}_{t-1};\theta_0,\eta_{1/\sqrt{T}})}{f_t(y_t \mid \mathcal{Y}_{t-1};\theta_0,\eta_0)} = \frac{1}{\sqrt{T}}\sum_{t=1}^{T} g_t(y^t) - \frac{1}{2}\mathcal{I}(g) + o_p(1)$$

Easily proved using the same method utilized for the iid case in Van der Vaart (2000)

Semiparametric Estimation Theory: Asymptotic Bias of MLE

MLE of θ in the baseline model is asypmtotically linear:

$$\hat{\theta}_T^{MLE} = I_0^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^{T} \dot{l}_t(y_t) + o_p(1)$$

Theorem 7 (Asymptotic Bias of MLE)

Let $r \mapsto \eta_r$ be a local deviation from η_0 that satisfies the uniform DQM property with score sequence $\{g_t(y^t)\}_{t\in\mathbb{N}}$ that has positive asymptotic variance and satisfies

$$\tau = \operatorname*{plim}_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \dot{l}_t(y_t) g_t(y^t).$$

Then,

$$\sqrt{T}\left[\hat{\theta}_T^{MLE} - \theta_0\right] \overset{\eta_{1/\sqrt{T}}}{\Rightarrow} \mathcal{N}\left[\tau,\ I_0^{-1}\right].$$

Proof follows from Le Cam's third lemma and the MDS CLT

Semiparametric Estimation Theory: Mean-Preserving Deviations

Back

Assume finite-dimensional parameter θ is identified through the conditional moment condition:

$$\mathbb{E}^{\eta_r}\left[\psi_t(y^t;\theta) \mid \mathcal{Y}_{t-1}\right] = O_{q\times 1} \quad \forall r \geq 0$$

Call any local deviation $r \mapsto \eta_r$ satisfying the identification condition **Identified Deviations**:

Theorem 8 (Characterization of Identified Deviations)

Suppose $r \mapsto \eta_r$ is an identified local deviation, and that it satisfies uniform DQM with score sequence $\{g_t(y^t)\}_{t\in\mathbb{N}}$. If:

i) $f_t(\cdot;\theta_0,\eta_r) \to f_t(\cdot;\theta_0,\eta_0)$ pointwise as $r \to 0$, and

ii) $\mathbb{V}^{\eta_r}\left[\psi_t(y^t;\theta_0) \mid \mathcal{Y}_{t-1}\right] \to \mathbb{V}^{\eta_0}\left[\psi_t(y^t;\theta_0) \mid \mathcal{Y}_{t-1}\right] = V_t(\theta_0)$ as $r \to 0$ for each t,

then we have

$$\mathbb{E}\left[\psi_t(y^t;\theta_0)g_t(Y_t) \mid \mathcal{Y}_{t-1}\right] = O_{q\times 1} \quad \forall t. \tag{1}$$

The set $\eta\dot{\mathcal{P}}_P$. of all score sequences satisfying equation (1) is our nuisance tangent space

Semiparametric Estimation Theory: Efficient Score Function

- The time t efficient score $\tilde{l}_t(y_t)$ is the "orthogonal projection" of baseline score $\dot{l}_t(y_t)$ onto the orthocomplement of $\eta\dot{\mathcal{P}}_P$ ($=$ span of $1, \psi_t(y^t;\theta_0)$)

- Under assumptions that allow us to interchange differentiation and integration,

$$\tilde{l}_t(y_t) = \underbrace{-\mathbb{E}\left[\frac{\partial \psi_t(y^t;\theta_0)}{\partial \theta} \mid \mathcal{Y}_{t-1}\right]' V_t(\theta_0)^{-1}}_{\Lambda_t(\theta_0)'} \psi_t(y^t;\theta_0)$$

- Locally robust estimator will be

$$\sqrt{T}\left[\hat{\theta}_T^{LR} - \theta_0\right] = \tilde{I}_0^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^{T} \tilde{l}_t(y_t) + o_p(1)$$

where

$$\tilde{I}_0 = \operatorname*{plim}_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \Lambda_t(\theta_0)' V_t(\theta_0)^{-1} \Lambda_t(\theta_0)$$

Semiparametric Estimation Theory: Locally Robust Estimator

Theorem 9 (Local Robustness and Efficiency)

Let $r \mapsto \eta_r$ be an identified deviation that satisfies the uniform DQM with score function sequence $\{g_t(y^t)\} \in \eta\dot{\mathcal{P}}_P$ admitting positive asymptotic variance, and $h \in \mathbb{R}^k$. Then,

$$\sqrt{T}\left[\hat{\theta}_T^{LR} - \left(\theta_0 + \frac{h}{\sqrt{T}}\right)\right] \overset{\theta_0+h/\sqrt{T},\eta_{1/\sqrt{T}}}{\Rightarrow} \mathcal{N}\left[O_{k\times 1},\ \tilde{I}_0^{-1}\right].$$

Let $\mathcal{T}_T$ be a regular estimator of θ:

$$\sqrt{T}\left[\mathcal{T}_T - \left(\theta_0 + \frac{h}{\sqrt{T}}\right)\right] \overset{\theta_0+h/\sqrt{T},\eta_{1/\sqrt{T}}}{\Rightarrow} \mathcal{T}^*$$

for any $h \in \mathbb{R}^k$ and identified deviation that satisfies the uniform DQM. Then, there exists a probability distribution M such that

$$\mathcal{T}^* = \mathcal{N}\left[O_{k\times 1},\ \tilde{I}_0^{-1}\right] * M.$$

$\hat{\theta}_T^{LR}$ is the locally robust estimator of θ that minimizes the efficiency loss from relying on the identification condition as our only source of information

Back

Suppose we are interested in identified deviations $r \mapsto \eta_r$ that satisfy

$$\mathbb{E}^{\eta_r}\left[m(y_t, \cdots, y_{t-\tau}; \theta_0)\right] = O_{q\times 1} \quad \forall r \geq 0$$

for some fixed $\tau > 0$. Define $z_t = (y_t, \cdots, y_{t-\tau})$.

Theorem 10 (Characterization of Identified Deviations)

Suppose $r \mapsto \eta_r$ is an identified local deviation, and that it satisfies uniform DQM with score sequence $\{g_t(y^t)\}_{t\in\mathbb{N}}$. If:

i) $f_t(\cdot; \theta_0, \eta_r) \to f_t(\cdot; \theta_0, \eta_0)$ pointwise as $r \to 0$, and

ii) $\mathbb{V}^{\eta_r}\left[m(z_t; \theta_0)\right] \to \mathbb{V}^{\eta_0}\left[m(z_t; \theta_0)\right]$ as $r \to 0$ for each t,

then we have

$$\mathbb{E}\left[m(z_{t+k}; \theta_0) g_t(y^t)\right] = O_{q\times 1} \quad \forall t, k \geq 0.$$

Semiparametric Estimation Theory: Unconditional Moment Conditions

- Again, the time t efficient score $\tilde{l}_t(y_t)$ is the "orthogonal projection" of $\dot{l}_t(y_t)$ onto the orthocomplement of $\eta\dot{\mathcal{P}}_P$ ($=$ span of $1, m(z_t;\theta_0)$)

- The efficient score becomes

$$\tilde{l}_t(y_t) = \Lambda_t(\theta_0)' \left[m(z_t;\theta_0) - \mathbb{E}_{t-1}\left[m(z_t;\theta_0) \right] \right],$$

  where

$$V_t(\theta_0) = \mathrm{Var}_{t-1}\left[m(z_t;\theta_0) \right]$$
$$\Lambda_t(\theta_0)' = \mathrm{Cov}_{t-1}\left[\dot{l}_t(y_t),\ m(z_t;\theta_0) \right] V_t(\theta_0)^{-1}$$

- As before, the locally robust estimator and efficient information will be

$$\sqrt{T}\left[\hat{\theta}_T^{LR} - \theta_0 \right] = \tilde{I}_0^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^{T} \tilde{l}_t(y_t) + o_p(1),$$

$$\tilde{I}_0 = \operatorname*{plim}_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \Lambda_t(\theta_0)' V_t(\theta_0)^{-1} \Lambda_t(\theta_0).$$

Semiparametric Estimation Theory: Unconditional Moment Conditions

Back

We require stronger assumptions on dependence:

- $\{m(z_t;\theta_0)\}_{t\in\mathbb{Z}}$ has bounded $2+\Delta$-moments for some $\Delta > 0$.
- There exists an α-mixing process $\{\epsilon_t\}_{t\in\mathbb{Z}}$ on which $\{m(z_t;\theta_0)\}_{t\in\mathbb{Z}}$ is L^2-NED with constants d_t^m that satisfy $\left(\mathbb{E}|m(z_t;\theta_0)|^{2+\Delta}\right)^{\frac{1}{2+\Delta}} \leq d_t^m$.
- The nuisance tangent space $\eta\dot{\mathcal{P}}_P$ contains sequences $\{G_t\}_{t\in\mathbb{N}}$ such that:
  i) $\{G_t\}_{t\in\mathbb{N}}$ is a martingale with respect to the filtration generated by $\{y_t\}_{t\in\mathbb{Z}}$ that has bounded $2+\Delta$-moments,
  ii) $\mathbb{E}[m(z_t;\theta_0)G_t] = 0$ for any $t \geq 0$, and
  iii) $\{G_t\}_{t\in\mathbb{N}}$ is L^2-NED on $\{\epsilon_t\}_{t\in\mathbb{Z}}$ with constants d_t^g that satisfy $\left(\mathbb{E}|G_t|^{2+\Delta}\right)^{\frac{1}{2+\Delta}} \leq d_t^g$.

Under these assumptions, for any $\{g_t(y^t)\}_{t\in\mathbb{N}}$ we can conclude that

$$\frac{1}{T}\sum_{t=1}^{T} m(z_t;\theta_0)g_t(y^t) \xrightarrow{p} O_{q\times 1}$$

by theorems 17.5, 17.9 and 19.11 of Davidson (1994).

Semiparametric Estimation Theory: Unconditional Moment Conditions

Theorem 11 (Local Robustness and Efficiency)

Let $r \mapsto \eta_r$ be an identified deviation that satisfies the uniform DQM with score function sequence $\{g_t(y^t)\} \in \eta\dot{\mathcal{P}}_P$ admitting positive asymptotic variance, and $h \in \mathbb{R}^k$. Then,

$$\sqrt{T}\left[\hat{\theta}_T^{LR} - \left(\theta_0 + \frac{h}{\sqrt{T}}\right)\right] \overset{\theta_0 + h/\sqrt{T}, \eta_{1/\sqrt{T}}}{\Rightarrow} \mathcal{N}\left[O_{k\times 1},\ \tilde{I}_0^{-1}\right].$$

Let $\mathcal{T}_T$ be a regular estimator of θ:

$$\sqrt{T}\left[\mathcal{T}_T - \left(\theta_0 + \frac{h}{\sqrt{T}}\right)\right] \overset{\theta_0 + h/\sqrt{T}, \eta_{1/\sqrt{T}}}{\Rightarrow} \mathcal{T}^*$$

for any $h \in \mathbb{R}^k$ and identified deviation that satisfies the uniform DQM. Then, there exists a probability distribution M such that

$$\mathcal{T}^* = \mathcal{N}\left[O_{k\times 1},\ \tilde{I}_0^{-1}\right] * M.$$

Proof once again follows from the MDS CLT and Le Cam's third lemma, coupled with the LLN result above that follows from the NED assumption

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- Now consider a local deviation under a autoregressive distributed lag (ARDL) specification:

$$
\begin{aligned}
x_t &= \rho x_{t-1} + \sigma_x \eta_t \\
y_t &= \phi y_{t-1} + \gamma x_{t-1} + \frac{c}{\sqrt{T}} \left(\omega_t - \frac{\mu}{1-\beta} \right) x_{t-1} + e^{\frac{1}{2}\omega_t} \epsilon_t \\
\omega_t &= \mu + \beta \omega_{t-1} + \sigma_\omega v_t, \quad (\epsilon_t, v_t, \eta_t) \overset{iid}{\sim} \mathit{StdNormal}
\end{aligned}
$$

- x_t is an exogenous variable that unilaterally Granger causes y_t

- Similar results obtain for local deviations from the ARDL model under parameters $\sigma_x^2 = \frac{e^{\mu/(1-\beta)+0.5\sigma_\omega^2}}{1-\phi^2}$, $\phi = 0.9$, $\gamma = 0.5$, $\mu = 0$, $\beta = 0.95$, $\sigma_\omega^2 = 0.5^2$

More Simulation Results for Augmented AR-SV Model

	$\left\|\hat{\gamma}_T^{LR} - \mathbb{E}[\gamma \mid \mathcal{Y}_T]\right\|$		$\lvert\gamma - \mathbb{E}[\gamma \mid \mathcal{Y}_T]\rvert$		$\frac{sd(\gamma \mid \mathcal{Y}_T)}{sd(\hat{\gamma}_T^{LR})}$		Coverage Probability	
T	Baseline	Aug	Baseline	Aug	Baseline	Aug	Baseline	Aug
100	0.0261	0.0846	0.0675	0.0844	0.950	2.362	0.53	0.87
250	0.0186	0.0238	0.0471	0.0343	0.902	1.470	0.42	0.84
500	0.0162	0.0096	0.0346	0.0222	0.868	1.253	0.37	0.86

Table: Monte Carlo Averages for $c = -1$, $\rho = 0$

	$\left\|\hat{\gamma}_T^{LR} - \mathbb{E}[\gamma \mid \mathcal{Y}_T]\right\|$		$\lvert\gamma - \mathbb{E}[\gamma \mid \mathcal{Y}_T]\rvert$		$\frac{sd(\gamma \mid \mathcal{Y}_T)}{sd(\hat{\gamma}_T^{LR})}$		Coverage Probability	
T	Baseline	Aug	Baseline	Aug	Baseline	Aug	Baseline	Aug
100	0.0276	0.0669	0.0755	0.0808	0.922	2.050	0.67	0.87
250	0.0223	0.0250	0.0469	0.0364	0.893	1.431	0.62	0.92
500	0.0172	0.0102	0.0334	0.0261	0.870	1.220	0.53	0.87

Table: Monte Carlo Averages for $c = -1$, $\rho = 0.9$